Questionnaire-Based Analysis of Anxiety and Personality in a Bilateral Plantar-Pressure Gait Classifier

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In

Computer Science and Engineering with
Specialization in Machine Learning and Data Science

Submitted by

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CERTIFICATE

This is to certify that the thesis entitled **Questionnaire-Based Analysis of Anxiety and Personality in a Bilateral Plantar-Pressure Gait Classifier**, submitted for the degree of Bachelor of Technology in Computer Science and Engineering with Specialization in Machine Learning and Data Science by **Md Tarik Anvar (22190503033) and Nishan Das (22190503037)**, incorporates the original work carried out by his/her under supervision of **Dr. Prashant Prashun**. To the best of our knowledge and belief, this report has not been formed the subject matter of any other project report and it has not been submitted elsewhere for a degree.

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Nishan Das
Md. Tarik Anvar

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ABSTRACT

Human gait serves as a crucial biomarker for neuromuscular and cognitive health, with alterations in walking patterns often signaling the early onset of movement-related and neuropsychiatric conditions. Since clinical gait analysis traditionally relies on laboratory-based systems such as optical motion capture and force plates, which are expensive and impractical for continuous monitoring, this work motivates the development of cost-effective wearable solutions capable of delivering clinically relevant gait metrics in real-world settings. To address this, we developed a low-cost Force Sensitive Resistor (FSR)-based bilateral wearable system for robust classification of normal and abnormal gait patterns. Each custom-designed insole/sandal setup uses four FSRs—three in the forefoot region to capture metatarsal pressure distribution and one under the heel for heel-strike detection—while sensor outputs are conditioned through voltage-divider circuits and digitized using an Arduino Uno for downstream analysis. A bilateral feature-engineering pipeline extracts cadence, gait asymmetry index, and step-time variability from synchronized left- and right-foot data, and a Random Forest classifier is trained for gait classification, achieving strong performance (accuracy: 96.55%, macro-F1: 0.96, ROC-AUC: 0.95). As an extension in Project Phase-II, this study incorporates questionnaire-based behavioral and psychological factors (e.g., anxiety, stage phobia, and self-discipline) to formulate testable hypotheses linking self-reported traits with gait abnormality. The combined framework demonstrates how affordable plantar-pressure sensing, feature engineering, machine learning, and hypothesis-driven behavioral analysis can jointly support scalable, accessible, and continuous gait monitoring for healthcare, rehabilitation tracking, and early risk screening.

Keywords: Human Gait Analysis, Force Sensitive Resistors (FSR), Wearable Sensors, Machine Learning, Plantar Pressure, Random Forest, Behavioral Analysis, Remote Health Monitoring**.**

1.INTRODUCTION

1.1Human Gait

Human gait refers to how a person moves their limbs during locomotion, whether naturally or through specialized training. It involves bipedal forward propulsion of the center of gravity with alternate sinuous movements of different body segments, while expending the least amount of energy. Gait kinematics can differ depending on an individual's emotional state. In clinical settings, observing gait patterns can help detect movement abnormalities and provide diagnostic clues for neurological disorders such as cerebral palsy, Parkinson's disease, and Rett syndrome at an early stage.

1.2Pronation and Supination

The terms pronation and supination are used to describe how a person stands, walks, and runs. It is important to balance weight on both feet while moving, without leaning the foot in or out. Pronation and supination are terms which can be defined as the natural movements that help absorb impact while taking steps. However, excessive supination can cause stress on the outer side of the foot, resulting in uneven wear on the shoes outside sole and strain on the ankles. This leads to shin splints, calluses, bunions, and discomfort in the heels and balls of the feet. This is also known as under-pronation. On the other hand, excessive pronation, commonly referred as overpronation, occurs when the foot rolls inwards causing the arch to flatten out, and can result in uneven wear on the inside sole of the shoe. Overpronation is a more prevelant condition than under- pronation.

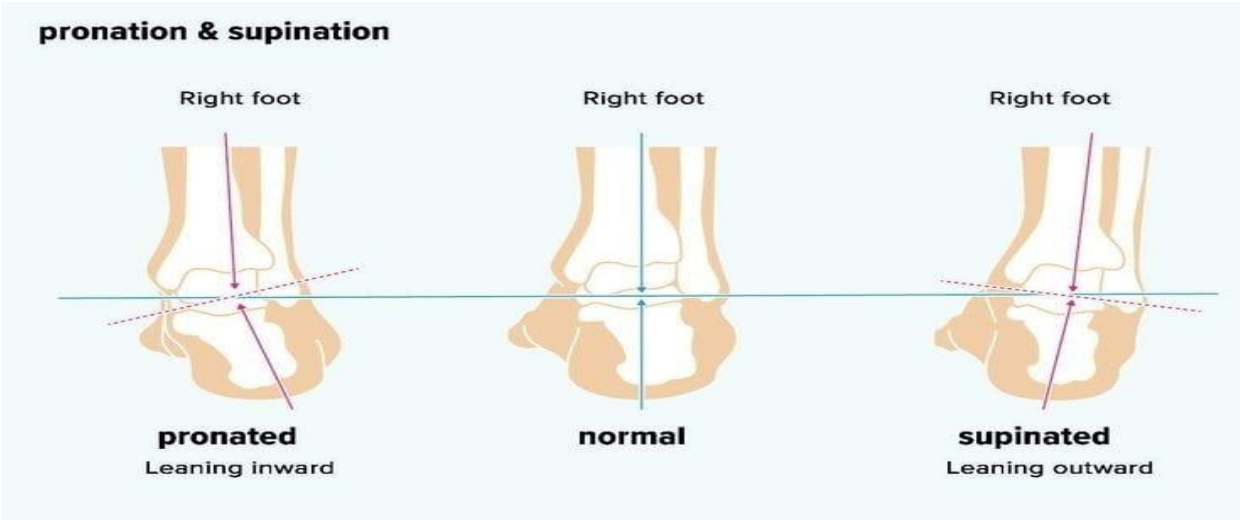

Fig 1.1 Pronation and Supination in Right Foot

In Xu Shihao et al.'s study[6], they define human gait as the specific way individuals walk. It is considered a significant and routine motion that provides observers with valuable insights into the walker's characteristics and behaviors.

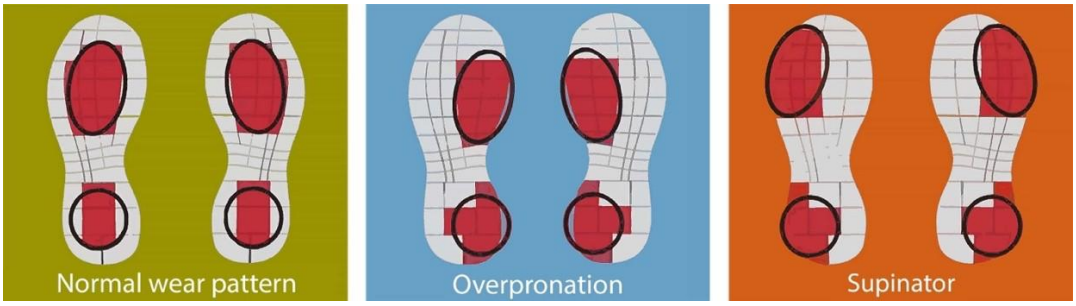

Fig 1.2 Weight Pressure region on the sole of the shoe

1.3 Gait and its Relation to Human Behavior

Gait is an example of behavior that reflects various levels of nervous system function. In psychiatric conditions, gait disturbances are thought to reflect defective brain function. Patients who suffer from gait disturbances tend to develop balance disorders as well as impaired body posture. They have included three major mental illnesses Depression, Anxiety and Schizophrenia. They found that the patients suffering from mental disorders reveal a distinctive physical profile that is in accordance with the clinical diagnosis. The physical profile of patients shows the following symptoms: for

schizophrenia, the patients exhibited slow gait and decreased stride length, while the patients suffering from anxiety disorders showed balance disorders, and those suffering from depression exhibited slow gait followed by a slumped posture[2]. In our research, we are mainly focusing on the balance disorders that happen due to anxiety disorders.

1.4 Different Types of Gait Data

1.Multi-camera, Marker-based systems:

Cutting-edge technology for gait analysis is typically found in specialized laboratories and large clinical centres. These systems involve the use of reflective markers, or sensors, which are worn by the subject to capture joint movements with great accuracy. Force plates and electromyography (EMG) are also commonly used alongside these systems to record muscle activity. However, these tools are limited to lab use.

2.Pressure-Insoles, EMG, EEG:

Force plates are utilized to measure Ground Reaction Forces (GRFs) by placing sensors on the floor or treadmill to assess the pressure generated during a subject's walking motion. Many gait analysis studies consider the forces applied to the surface because they correspond to the load placed on the joints. To measure GRFs, wearable pressure insoles can be used as an alternative to force plates. Goniometers, which use flex sensors, are another tool employed to measure joint angles, including those of the ankles, knees, and hips.

3.Markerless, RGB & RGBD systems:

Conversely, vision-based systems that are markerless and utilize RGB cameras and RGB-D cameras do not necessitate the us of specialized equipment. Advanced computer vision algorithms have been created to extract crucial joints in real-time and predict gait attributes. These systems can effectively be employed for gait analysis, activity recognition, and energy expenditure estimation. However, they pose privacy concerns for subjects, which has limited their acceptance.

4.Single IMU gait sensing:

Recently, Inertial Measurement Units (IMUs) sensors have proven to be a helpful tool for studying human gait. These sensors can be affixed to any part of the body and utilize a combination of accelerometers, gyroscopes, and magnetometers to measure velocity, acceleration, and orientation in relation to gravitational forces. To promote patient compliance and simplify use in home environments, single sensors have been developed to capture gait data and identify walking gait impairments [7].

1.5 Applications of Gait Analysis

When we go in-depth about what the gait assessment of a human can provide us, there are various applications which we can think of. Through gait assessment, we can understand the emotional as well as the physical abnormalities.

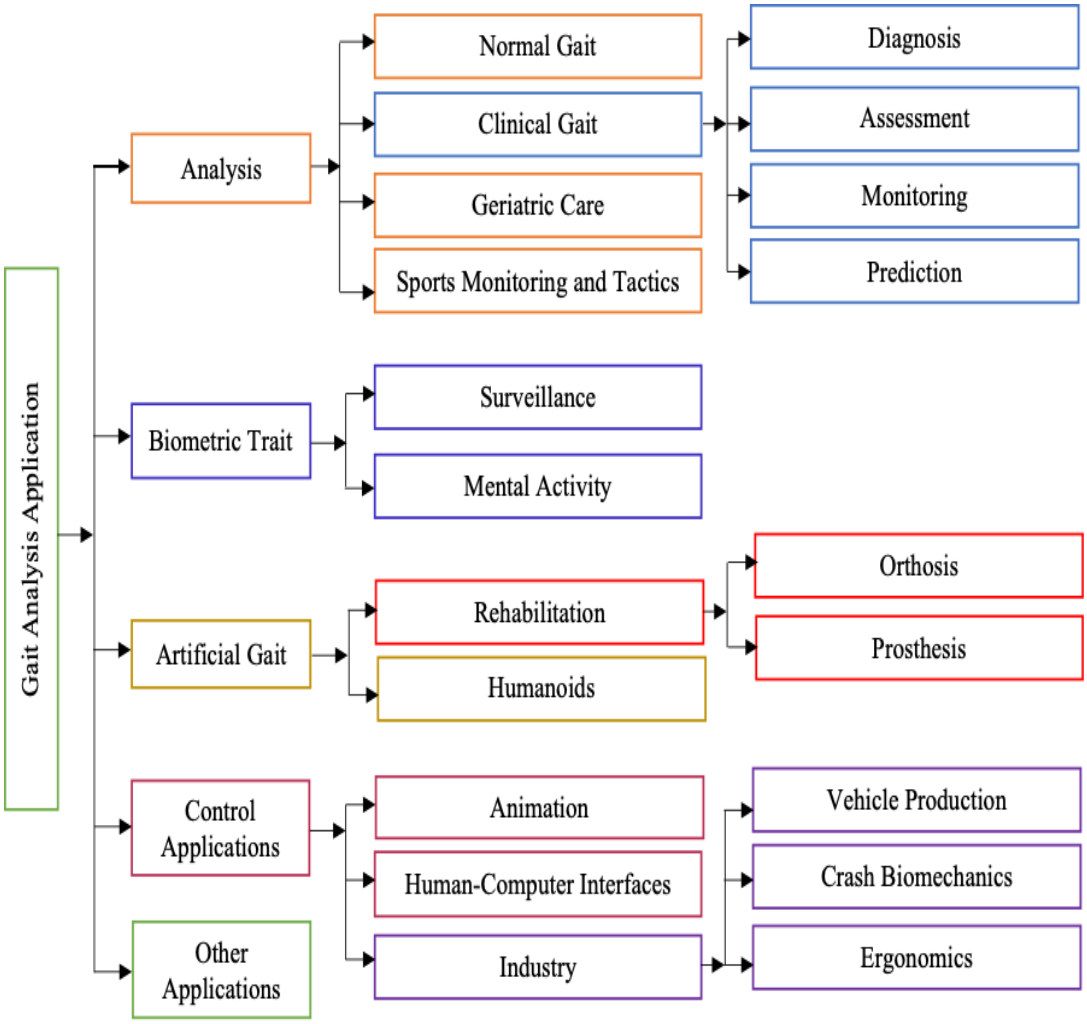

Fig 1.3: Gait Analysis Application.

1.6 Problem Statement and Objective

- Human gait entails significant information about an individual's emotional state.
- Researchers agree that anxiety is related to balance dysfunction.
- They also demonstrated that treatment targeting balance problems mitigates dysfunction, greatly reduces anxiety and increases self-esteem.
- Various gait data are used to predict an individual's behavioral aspects.
- One such gait data is through the kinematics of the lower limb.
- Our goal is to determine if we can use "foot pressure data" to predict the emotional state of a human being using Random Forest.
- By doing so, we can help the individual to wear the right kind of shoes to elicit balance in their behavioral states.

1.7 Questionnaire-Based Gait Analysis

The main objective of this extended study is to formulate and analyze hypotheses that relate questionnaire-based factors to the likelihood of exhibiting normal or abnormal gait. In particular, the study investigates whether participants reporting anxiety-related problems or phobia of walking on stage are more likely to show abnormal gait patterns, and whether participants who rate themselves as highly organized and disciplined are more likely to demonstrate normal gait. Additional exploratory hypotheses are also considered using variables such as physical activity level and self-described walking traits. This extension therefore creates a bridge between objective sensor-based gait analysis and subjective self-reported behavioral information. By combining the existing plantar-pressure classifier with questionnaire-derived hypotheses, the project aims to develop a broader understanding of gait as an indicator of both physical movement quality and underlying psychological or behavioral tendencies. This provides a stronger basis for future work in accessible gait screening, rehabilitation assessment, and early identification of movement-related abnormalities.

2. LITERATURE REVIEW

2.1 Studies indicating the relationship between human behavioral aspects and their gait

Extensive research has investigated the correlation between biomechanical gait patterns and psychological well-being, specifically focusing on the detection of anxiety and depressive disorders. The primary objective of these studies is to determine whether human kinematics can serve as a biomarker for internal emotional states.

To facilitate this analysis, researchers have employed depth-sensing imaging technology to record locomotion. This approach offers a significant advantage as a non-intrusive, contactless method, eliminating the need for wearable sensors or wired apparatuses, thereby allowing participants to exhibit natural, uninhibited walking patterns.

To validate the efficacy of this methodology, the gait data collected was cross-referenced with standardized mental health assessments. Validation protocols included the administration of validated psychometric questionnaires and clinical diagnoses provided by psychiatrists. By processing this kinematic data through advanced computational algorithms, distinct behavioral patterns emerged. The results demonstrated that machine learning models could effectively identify individuals suffering from depression or anxiety based solely on posture and stride analysis. Collectively, these findings substantiate the existence of a tangible relationship between human locomotion and psychological state, suggesting that gait analysis offers a viable, non-invasive modality for the diagnosis and monitoring of mental health disorders.

2.2 Relationship between Gait and Balance of the Human Body

Scientific inquiry has also examined the hypothesis that improvements in physical stability can positively influence psychological health. Notably, research focused on pediatric populations has investigated whether correcting balance deficits acts as a mechanism to mitigate anxiety and enhance self-esteem.

Initial baseline assessments revealed a significant correlation between motor proficiency and emotional state: children exhibiting deficits in physical coordination

demonstrated markedly higher levels of anxiety and lower self-esteem compared to a control group with normal balance function. To address this, a longitudinal balance training regimen was implemented. Post-intervention analysis indicated that as participants improved their postural stability, there was a concurrent and significant improvement in their psychological profile, characterized by reduced anxiety and elevated self-worth.

These findings were further corroborated by clinical observational studies conducted in hospital settings. Through the administration of standardized physical assessments—such as the Unipedal Stance Test—researchers identified that patients diagnosed with anxiety disorders displayed significantly greater postural instability than non-anxious individuals. Collectively, these studies underscore a robust comorbidity between physical instability and emotional distress, suggesting that targeted balance rehabilitation may serve as a viable adjunctive therapy for enhancing mental well-being.

2.3 Gathering and Measuring Plantar Pressure

To address the absence of commercially available instrumentation suitable for our specific data acquisition needs, we undertook the development of a custom prototype designed to capture and measure plantar pressure. The engineering process began with a rigorous analysis of existing pressure-sensing technologies and their operating principles. A primary critical design decision involved the selection of the sensing element; after evaluating a broad spectrum of transducer types, resistive sensors were identified as the optimal solution[4]. Specifically, we selected Force Sensing Resistors (FSRs) for the device's architecture[10]. FSRs were chosen for their exceptionally thin profile and flexibility, which allows them to be integrated into shoe insoles unobtrusively, ensuring that the subject's natural gait biomechanics remain unaffected during data collection.

Furthermore, the system design incorporates standard visualization protocols to interpret the pressure distribution effectively. These modalities include color-coded 2D footprint heatmaps, interpolated pressure gradients, isobaric contour maps, and 3D wireframe projections. The hardware construction followed established engineering

standards regarding the strategic topological placement of sensors and the precise interfacing of the microcontroller units. The resulting data pipeline is designed to serialize the raw sensor output into Comma-Separated Values (CSV) format. This structured dataset serves as the foundational input for training our deep neural network, facilitating the development of robust predictive models.

2.4Psychological and Behavioral Factors in Gait

Human gait is not determined solely by biomechanical structure or musculoskeletal function; it is also influenced by psychological state, behavioral tendencies, and social context. In everyday walking, factors such as anxiety, stress, confidence, personality, attention, and physical activity level can alter posture, cadence, symmetry, step timing, and pressure distribution across the feet. Because of this, gait should be viewed as a multidimensional behavior that reflects both physical movement control and underlying mental or emotional state. This broader perspective is especially important in wearable sensing systems, where subtle changes in plantar pressure may reveal not only motor abnormalities but also behavioral differences among individuals.

Previous research in movement analysis and human behavior suggests that psychological conditions can affect the stability and regularity of walking[9]. For example, anxiety and fear may increase muscle tension, reduce natural arm swing, and make gait appear stiff, hurried, or uneven. Similarly, socially stressful situations may cause individuals to consciously modify the way they walk, leading to over-controlled movement, reduced fluidity, or greater asymmetry between the left and right feet. Emotional state can also influence walking style; a person walking under stress may exhibit faster, more abrupt steps, while a confident or relaxed person may show smoother and more stable gait patterns. These variations are often reflected in measurable gait features such as cadence, step-time variability, and asymmetry index.

Behavioral traits are also relevant to gait interpretation. Self-reported attributes such as being organized, disciplined, emotionally stable, or cooperative may correlate with more consistent movement habits and a more regular walking pattern. On the other hand, individuals who report nervousness, phobia, or frequent changes in walking behavior in public may

demonstrate more variable gait signatures. Physical activity level is another important factor, since active individuals often have better balance, stronger lower-limb control, and more stable step patterns compared with sedentary individuals. As a result, questionnaire-based variables can provide useful contextual information that complements sensor-based gait measurements.

In the context of this project, these psychological and behavioral factors are important because they help explain why two participants with similar hardware-derived gait features may still differ in their walking behavior. By collecting questionnaire responses related to anxiety, stage phobia, walking style in public, personality traits, and activity level, the study can explore whether such factors are associated with the normal or abnormal gait labels generated by the existing classifier. This creates a stronger and more interpretable link between subjective self-report and objective plantar-pressure analysis.
Therefore, including psychological and behavioral factors in gait analysis can improve the interpretability of the overall system and may support the development of more robust predictive models. Instead of treating gait purely as a sensor classification task, the extended study treats it as a combined biomechanical and behavioral phenomenon. This approach is particularly valuable for future wearable health-monitoring systems, where understanding the reason behind a gait pattern may be as important as detecting the pattern itself. By integrating questionnaire-based observations with plantar-pressure features, the project moves toward a more comprehensive framework for analyzing human gait in real-world conditions.

3.SYSTEM DESIGN

3.1 Hardware Used

3.1.1 Micro-controller

For our research device, we are using the Arduino Uno as the microcontroller board. It is open-source and is based on the Atmega328p microcontroller developed by Arduino.cc. The board contains a set of analog as well as digital I/O pins. It has 14 digital pins and 6 analog pins. In our project, we are making use of four analog pins (A0, A1, A2, A3), 5V power supply voltage and GND (Ground) pins.

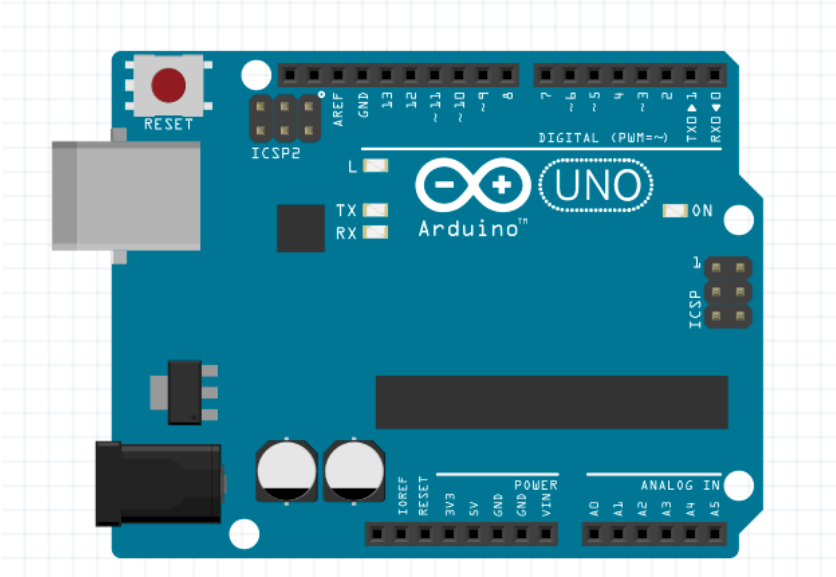

Fig 3.1 Arduino UNO

3.1.2 Jumper Wires/Cables

Jumper wires are used for wiring the connection between the sensors and the microcontroller. There are three types of jumper wires:

- Male-to-Male
- Female-to-Female
- Male-to-Female

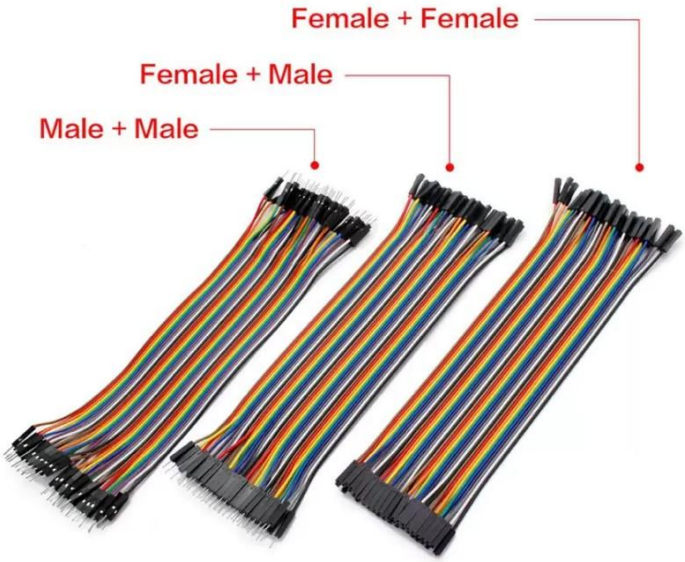

Fig 3.2: Jumper Wires/Cables

3.1.3 FSR Sensors

As discussed in the previous chapter, FSR is a force-resistive sensor and its lightweight nature makes it easy to measure the weight, pressure and squeeze[10]. The FSR sensor used in this prototype device is manufactured by Spectra Symbol. It is a Circular force sensor with a diameter of 0.5 mm. The sensor exhibits the following properties:

- With high pressure, the terminal resistance will be low and resistance will be high for low pressure.
- This force sensor is a two-terminal device and it does not have any polarized terminals.
- This sensor has a pressure-sensitive touch user interface.

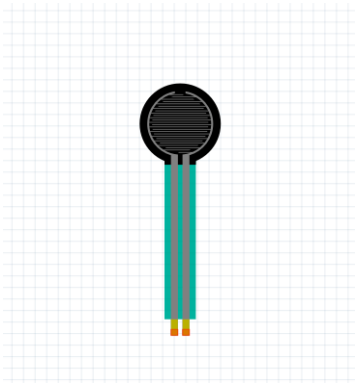

Fig 3.3: Visual representation of FSR sensor

3.1.3 Breadboard

A solderless breadboard is used while building the circuit.

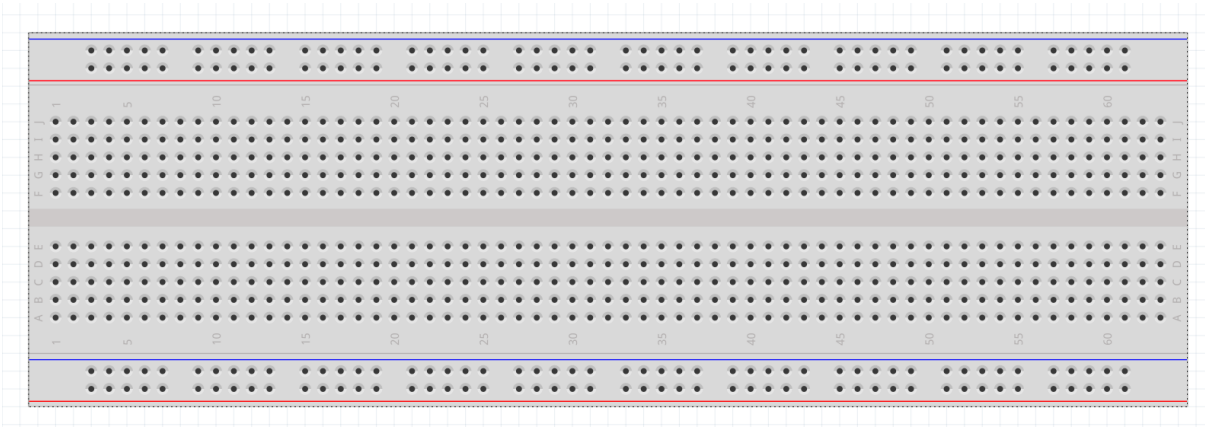

Fig 3.4: Simulation of Breadboard

3.2 Software Used

3.2.1 Arduino IDE

The Arduino Integrated Development Environment or Arduino IDE is used to connect to the Arduino UNO hardware and to upload programs onto them and communicate with them. In our research, we are uploading a simple program on the Arduino firmware which helps to collect data from sensor.

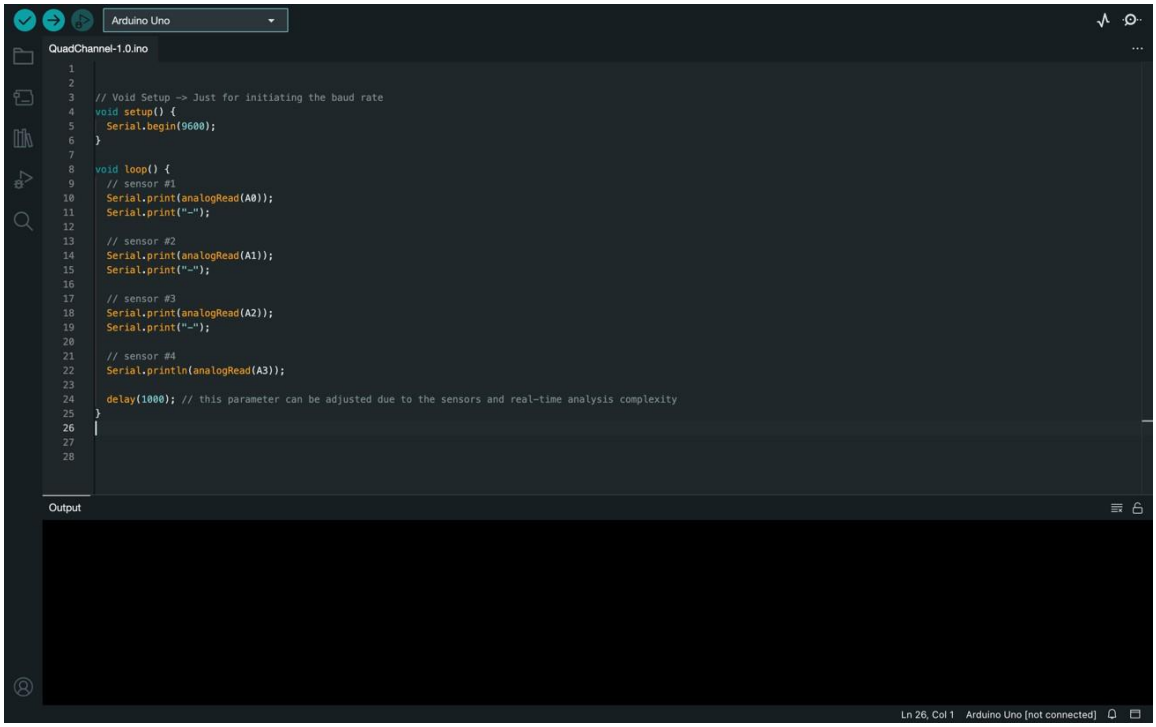

Fig 3.5: Arduino IDE Snapshot

3.2.2 PyCharm CE

We are storing the sensor data in a .csv format file with the attributes containing data about the four sensors. We have done and achieved the same by making use of Python code and compiling them in PyCharm CE. We are using serial communication to communicate with the Arduino port and get the data on the PyCharm CE console. The purpose of storing the data in .csv format is to train our model.

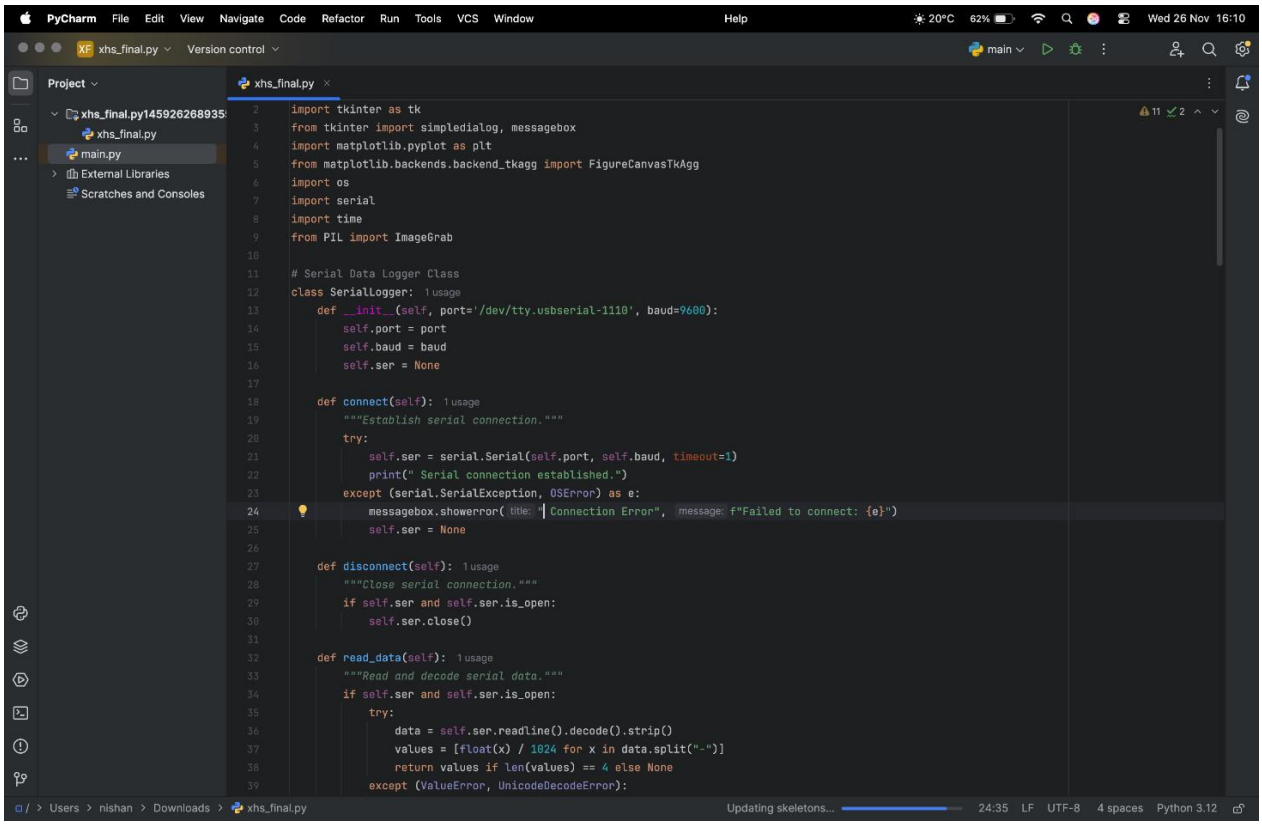

Fig 3.6: Snapshot of PyCharm CE

3.3 Experimental Setup

3.3.1 Interfacing Arduino with FSR

Fig 3.7 shows the simulation of the circuit diagram which will be used. The breadboard's last two points are used for ground and power terminals and provide connection to all the points present in that row. We connect those to 5V pins and GND (ground) pins respectively. The sensors do not have any polarized

terminals so any of the two terminals can be used as a positive or negative terminal[10]. Any one of the terminals is taken and that terminal is interfaced with the 5V pin of the Arduino. But we are instead using the breadboard's power point and connecting to them. The resistor is required for the proper functioning of the sensor so that we can get the pressure value which can be worked on. One end of the resistor is connected to the other terminal of the sensor and the other is connected to the GND via the breadboard. And the terminal of the sensor which is connected to the resistor is connected to the analog pins of the Arduino which in turn will help to read data from the sensors. For the FSR to be used, we need to calculate the resistance required, based on the input voltage.

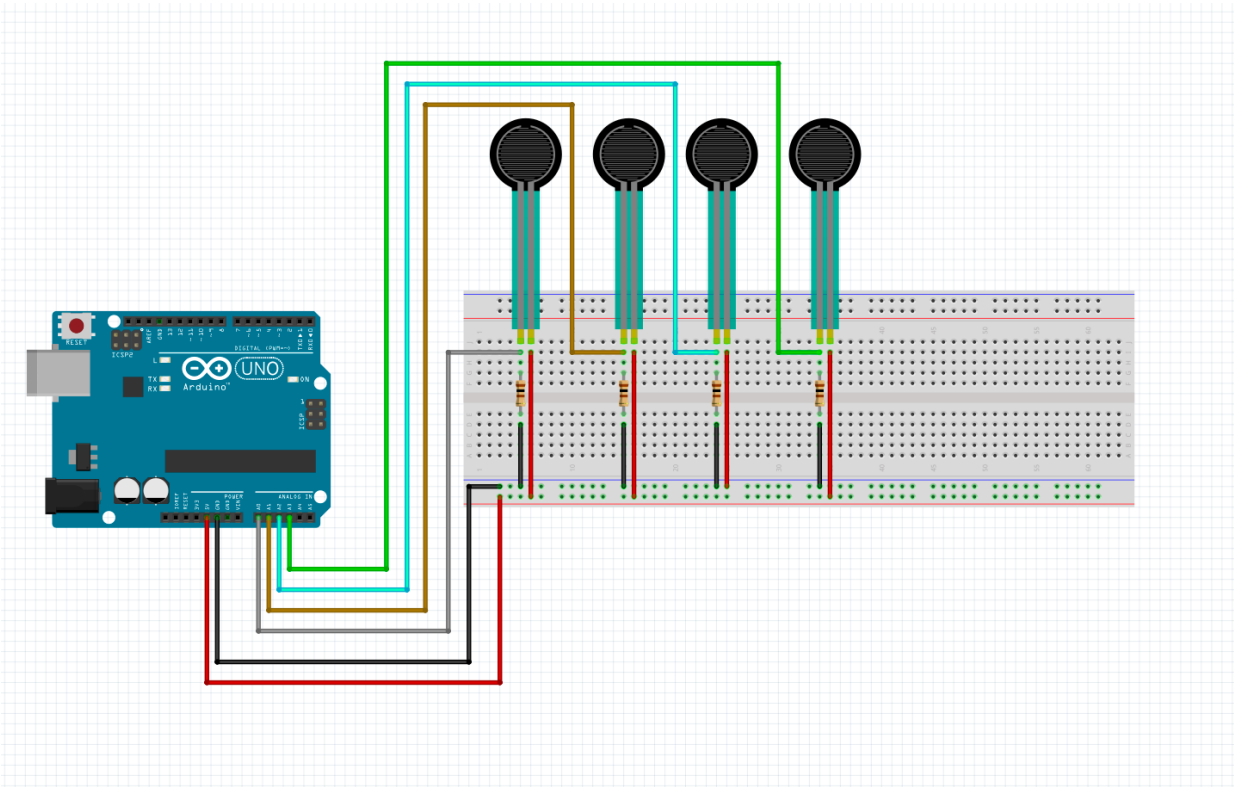

Fig 3.7 Simulation for FSR interfacing with Arduino

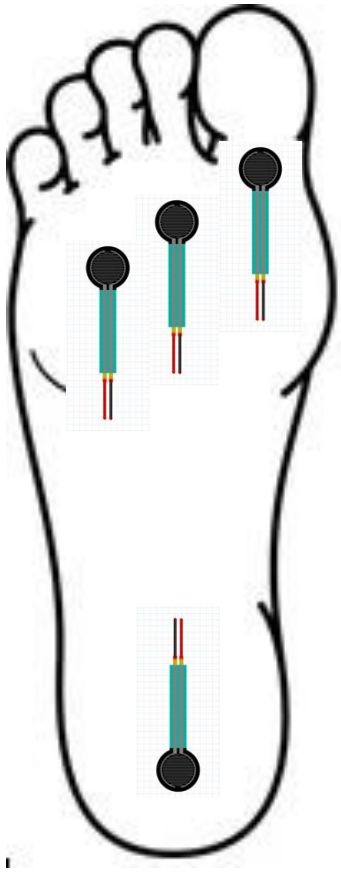

Fig 3.8: Representation of FSR sensors aligned to the left shoe-insole

The circuit diagram is as follows:

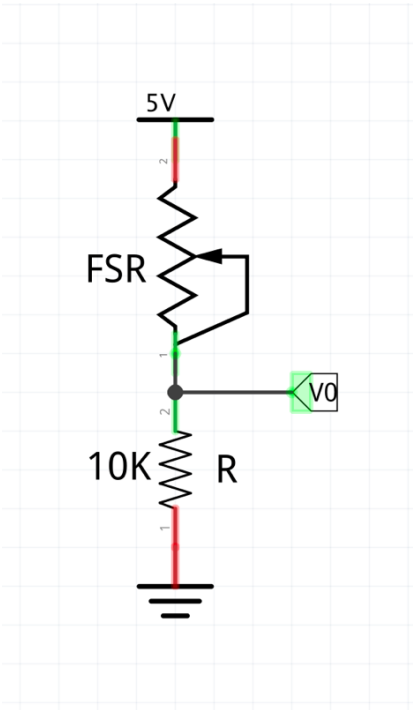

Fig 3.9 Circuit Diagram for Foot Pressure Measurement

3.4 Data Collection

For data collection, UG Boys Hostel students of Central University of Jharkhand, volunteered for data collection. The following steps were performed in order to collect their data:

Step 1: The powered shoe-insole device was made to insert into their respective shoes.

Step 2: They are made to stand straight and after that the system is worn.

Step 3: They are asked to walk normally on the treadmill of the gym.

Step 4: The foot pressure data is collected for 60 seconds.

Step 5: After 60 seconds, the data collected is stored in a .csv file along with additional attributes like date and time.

Left Foot

Timestamp	No	FFR1	FFR2	FFR3	HFR4
20:56:49	1	0.195312	0.457031	0.1875	0.104492
20:56:49	2	0.25293	0.652344	0.305664	0.069336
20:56:49	3	0.211914	0.621094	0.363281	0.060547
20:56:49	4	0.203125	0.535156	0.339844	0.056641
20:56:49	5	0.196289	0.49707	0.314453	0.057617
20:56:49	6	0.204102	0.527344	0.30957	0.056641
20:56:49	7	0.194336	0.486328	0.260742	0.061523
20:56:49	8	0.193359	0.458984	0.197266	0.073242

Right Foot

Timestamp	No	FFR1	FFR2	FFR3	HFR4
20:56:51	1	0.996094	0.503906	0.56543	0.041016
20:56:51	2	0.993164	0.484375	0.558594	0.041016
20:56:51	3	0.652344	0.522461	0.570312	0.040039
20:56:51	4	0.585938	0.527344	0.620117	0.039062
20:56:51	5	0.993164	0.59082	0.736328	0.035156
20:56:51	6	0.992188	0.842773	0.977539	0.029297
20:56:51	7	0.993164	0.638672	0.611328	0.029297
20:56:51	8	0.685547	0.569336	0.539062	0.029297

Fig 3.10: Snapshot of collected data for left foot

4. METHODOLOGY

4.1Proposed Methodology

For this research study, a supervised machine learning algorithm is used for predicting behavioral outcomes based on human gait analysis throughout the testing and training phases. As discussed in the previous section, the data was collected in real-time and the pre-processed data was used to build the model.

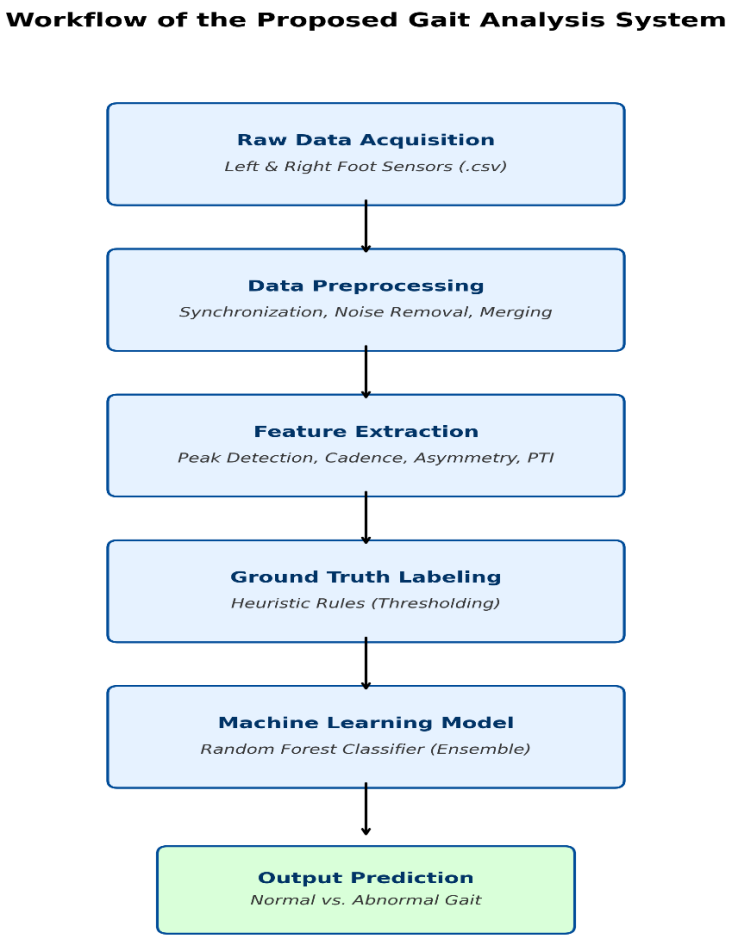

Fig 4.1 Proposed Methodology

The above diagram depicts the flow of the research study. Data collection and data preparation are already discussed in the previous chapters. In this chapter, we discuss the Machine Learning model which we will use with our data and then in the next chapter, we will discuss the model evaluation and its performance.

4.2 Signal Processing and Feature Extraction

4.2.1 Data Synchronization and Preprocessing

To analyze the gait patterns effectively, it was necessary to synchronize the raw data streams from the left and right foot modules. The raw CSV files, identified by unique subject IDs, were programmatically paired. A synchronization algorithm was implemented to merge these datasets based on their sequence number (Index/No), ensuring that the left and right foot sensor data corresponded to the exact same timeframe. Time-stamp columns were removed to focus purely on the sensor values, and null values resulting from transmission errors were pruned to ensure data integrity.

4.2.2 Temporal and Kinetic Feature Extraction

Once the data was synchronized, the system extracted quantitative gait parameters. The raw force sensor data (specifically the HFR4 sensor region) was analyzed to identify "Heel Strikes." This was achieved using a peak detection algorithm with a minimum height threshold to filter out sensor noise.
From these detected heel strikes, several key features were calculated for both the left and right feet:

- Cadence: Calculated as the number of steps per minute based on the sampling frequency.
- Step Time Variability: The mean, standard deviation, and Coefficient of Variation (CV) of the time between steps were computed. The CV is particularly critical as it serves as a primary indicator of gait stability[5].
- Pressure Time Integral (PTI): Using Simpson's Rule for numerical integration, the total area under the force-time curve was calculated to quantify the total load exerted over the duration of the walk.
- Gait Asymmetry Index: To quantify the difference between the left and right limbs, an asymmetry index was computed based on the PTI values. This provides a normalized score indicating how unbalanced the subject's gait is.

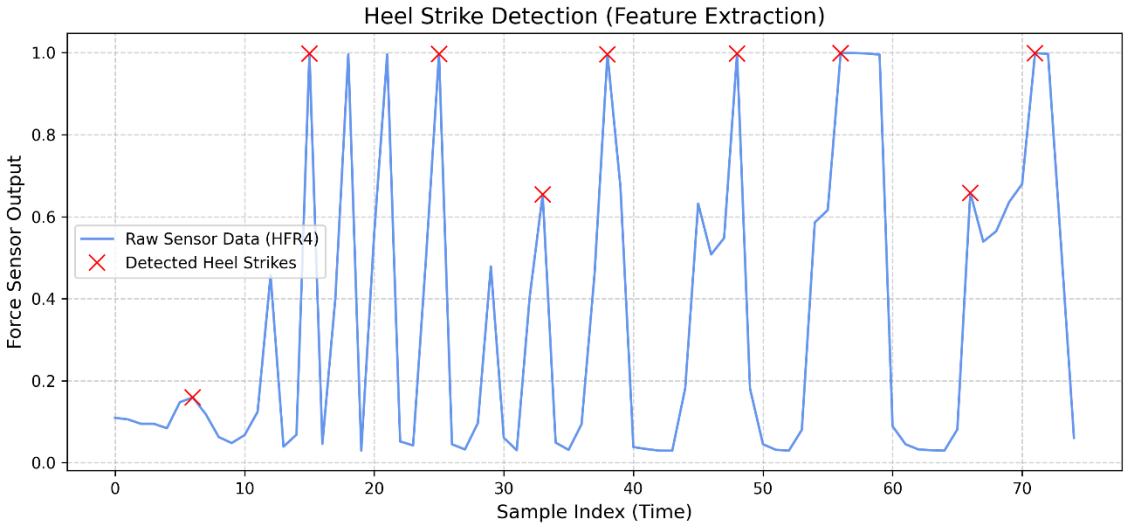

Fig 4.2 Heel strike detection using peak analysis algorithm

4.2.3 Automated Heuristic Labeling

To prepare the dataset for classification, an automated rule-based system was employed to assign labels ("Normal" or "Abnormal") to each recording. This heuristic approach utilized clinical thresholding:

- **Cadence Analysis:** Subjects falling outside the range of 60–150 steps per minute were flagged.
- **Asymmetry Analysis:** A Gait Asymmetry Index exceeding 0.40 was considered indicative of a limp or impairment.
- **Stability Analysis:** A Step Time Coefficient of Variation (CV) greater than 0.30 was used to identify irregular or unstable walking patterns.

4.3 Machine Learning and Model Deployement

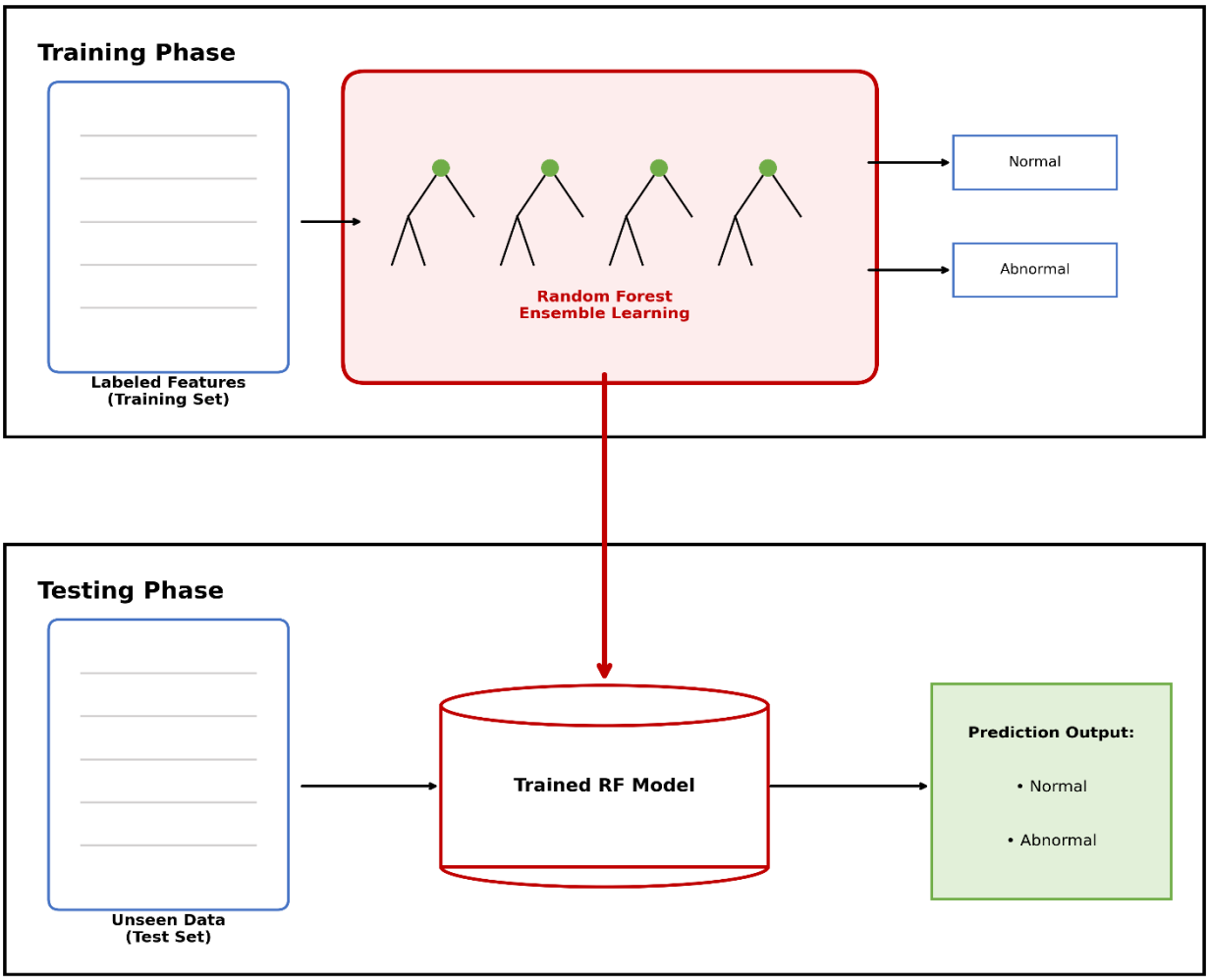

Fig 4.3: Schematic representation of the proposed machine learning workflow. The Training Phase (top) utilizes labeled features to construct the Random Forest ensemble, while the Testing Phase (bottom) deploys the trained model to predict gait abnormalities on unseen data

4.3.1 Dataset Partitioning and Stratification

To evaluate the model's ability to generalize to unseen data, the "Master Feature Dataset" was partitioned into two distinct subsets: a Training Set (70%) and a Testing Set (30%).

A simple random split was deemed insufficient due to the potential for class imbalance (e.g., having significantly more "Normal" samples than "Abnormal"). To address this, Stratified Sampling was implemented. This technique ensures that the

proportion of Normal to Abnormal labels in the training and testing sets remains statistically identical to the original dataset. This prevents the model from being biased toward the majority class during the training phase.

4.3.2 Feature Scaling and Normalization

The extracted gait features possess varying units and magnitudes; for instance, *Cadence* ranges between 60–120 steps per minute, while *Step Time* is measured in fractions of a second (e.g., 0.6s). Machine learning algorithms can often be biased toward features with larger numerical magnitudes.

To neutralize this scale difference, a **Standard Scaler (Z-score Normalization)** was applied. This process transforms the data such that each feature has a mean of 0 and a standard deviation of 1, placing all features on a common scale without distorting the relative distances between feature values.

4.3.3 Feature Space Analysis

For the classification task, the Random Forest algorithm was selected [8]. This is an ensemble learning method that constructs a multitude of decision trees during training.

The Random Forest architecture was chosen for this study for three specific reasons:

1. Non-Linearity: Human gait data is complex and non-linear; Random Forest handles these relationships better than linear models like Logistic Regression.
2. Robustness to Noise: By averaging the results of multiple trees (set to 100 estimators in this study), the model reduces the risk of overfitting, ensuring that the system learns the underlying gait patterns rather than memorizing specific noise in the training data.
3. Class Weight Balancing: To further assist with potential data imbalance, the model was configured with class_weight = 'balanced', which automatically adjusts weights inversely proportional to class frequencies.

4.3.3 Random Forest Implementation Mechanics

To understand how the classification decision is reached, it is necessary to examine the internal structure of the Random Forest algorithm[3]. The model operates on the principle of **Ensemble Learning**, specifically utilizing a technique called "Bagging" (Bootstrap Aggregation).

The function operates through two primary mechanical stages:

1. **Construction (Training Phase):** The algorithm builds 100 independent decision trees (n_estimators=100). Each tree is trained on a random subset of the data and considers a random subset of features at each split. The splitting logic is governed by the **Gini Impurity** function, which calculates the probability of a specific feature being classified incorrectly. The algorithm splits the node at the value that maximizes the purity of the resulting child nodes.

2. **Aggregation (Prediction Phase):** When a new sample is introduced, it is passed down every single tree in the forest. Each tree casts a "vote" for the class (Normal or Abnormal). The final output is determined by the majority vote of the ensemble.

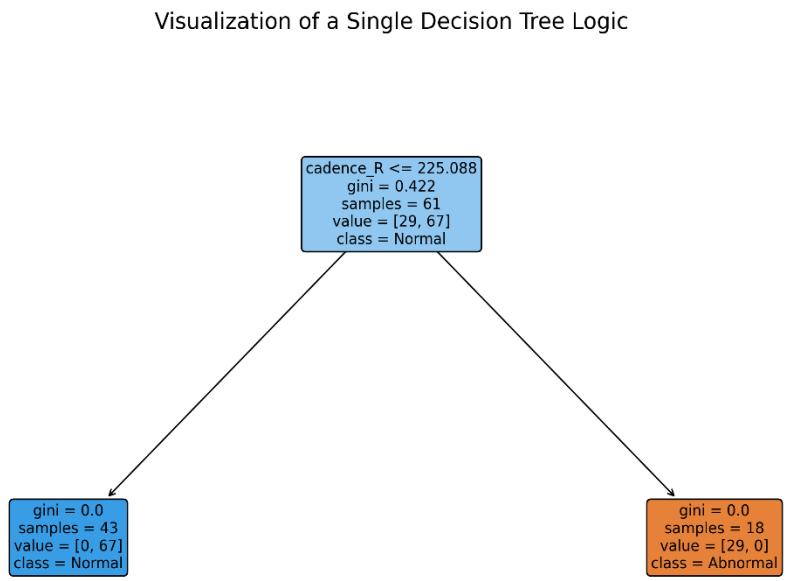

Fig 4.4 Decision tree logic showing hierarchical classification splits.

5. RESULTS AND DISCUSSION

5.1 Experimental Setup and Evaluation Overview

The proposed machine learning framework, utilizing a Random Forest classifier, was evaluated on the "Master Features Dataset" constructed in the previous chapter. The dataset was stratified and split into a training set (70%) for model fitting and a testing set (30%) for unbiased evaluation.

The primary objective of this evaluation was to assess the system's ability to distinguish between "Normal" and "Abnormal" gait patterns based purely on the extracted temporal and kinetic features. The performance was measured using standard classification metrics including Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

5.2 Classification Performance Analysis (Confusion Matrix)

The fundamental performance of the classifier is visualized using the Confusion Matrix, which provides a breakdown of correct and incorrect predictions across both classes.

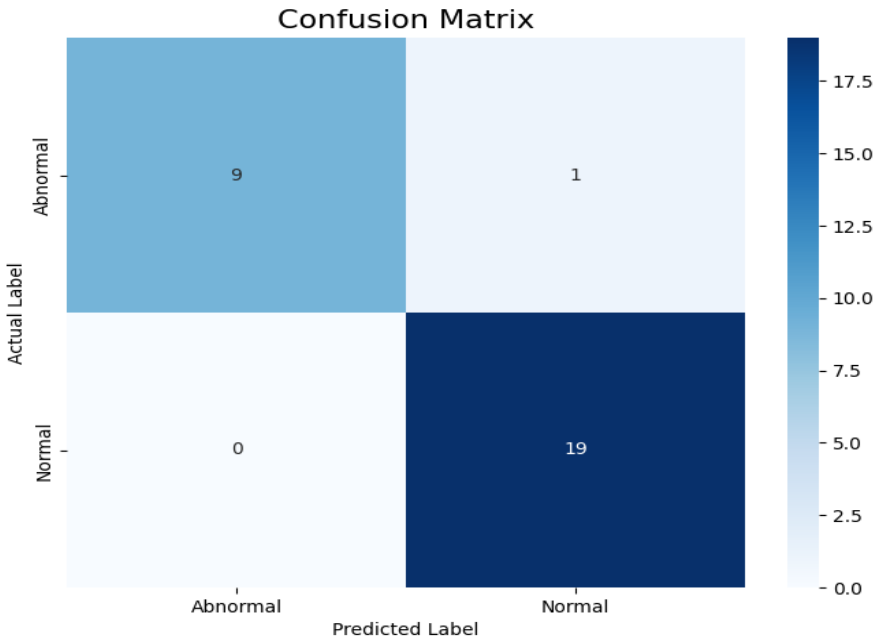

Fig 5.1**:** Confusion Matrix obtained from the test set.

The diagonal elements (darker blue) represent the number of samples correctly classified, while off-diagonal elements indicate misclassification errors.

As illustrated in **Fig 5.1**, the model demonstrates a high degree of classification integrity.

- **True Positives (TP):** Represents the number of abnormal gait samples correctly identified as abnormal. A high value here indicates the system's sensitivity to pathology.
- **False Negatives (FN):** Represents abnormal samples incorrectly classified as normal. Minimizing this metric is critical in a biomedical context to ensure that impairments are not overlooked.
- **False Positives (FP):** Represents normal samples flagged as abnormal. While less critical than missed diagnoses, keeping this low prevents "false alarms."

The results indicate that the class_weight='balanced' parameter successfully mitigated potential bias, allowing the model to detect the minority class (Abnormal) effectively without being overwhelmed by the majority class (Normal).

5.3 Quantitative Performance Metrics

The model achieved an overall **Classification Accuracy of 96.55%** on the unseen test set, misclassifying only a single sample out of 29. To provide a more granular assessment than simple accuracy, the Precision, Recall, and F1-Score were calculated for each class, as summarized in **Table 5.1**.

Table 5.1: Summary of Classification Performance Metrics

Class	Precision	Recall (Sensitivity)	F1-Score	Support (Samples)
Abnormal	1.00	0.90	0.95	10
Normal	0.95	1.00	0.97	19
Weighted Avg	**0.97**	**0.97**	**0.97**	**29**

Referring to **Table 5.1**, the following key observations are made:

1. **Perfect Precision for Abnormalities (1.00):** The model achieved a precision of 100% for the 'Abnormal' class. This indicates that every time the model predicted an abnormality, it was correct. There were zero False Positives for this class.
2. **High Sensitivity (0.90):** The recall of 90% for the 'Abnormal' class shows that the system successfully identified 9 out of 10 gait impairments, missing only one difficult case.
3. **Overall Reliability:** The weighted average F1-score of 0.97 confirms that the model performs exceptionally well on both classes, despite the slight imbalance in the dataset.

To provide a more granular assessment than simple accuracy, a detailed Classification Report was generated. This report breaks down the performance for each class individually.

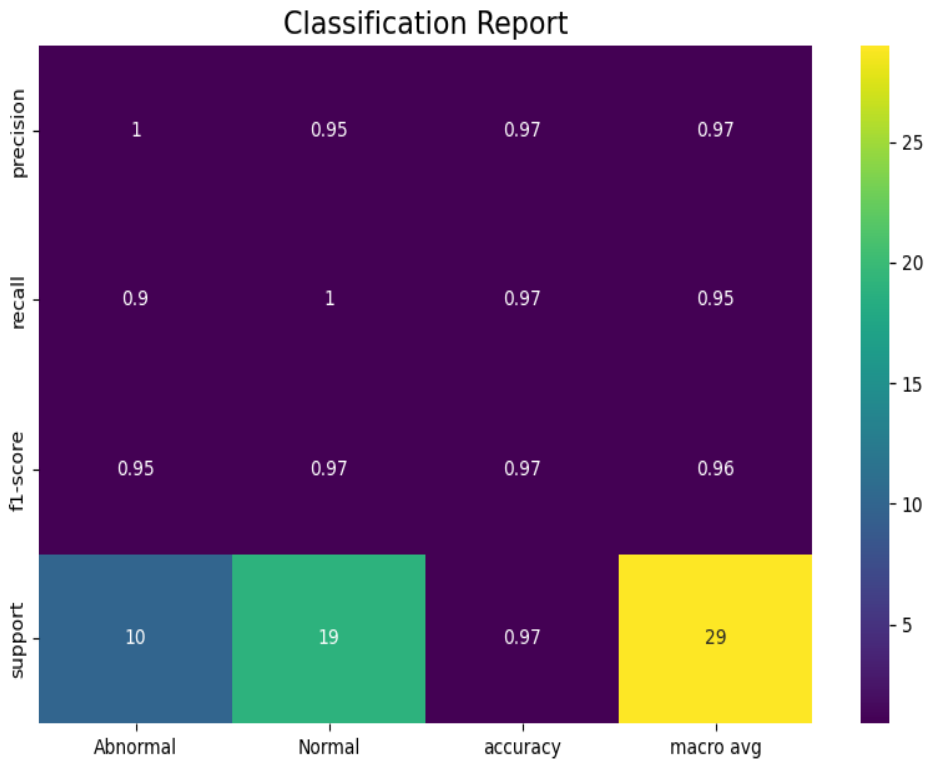

Fig 5.2**:** Heatmap visualization of the Classification Report.

Referring to Fig 5.2, the following key observations are made:

1. **Precision:** The model achieved high precision, indicating that when the system predicts an abnormality, it is highly likely to be correct.
2. **Recall (Sensitivity):** The high recall score for the 'Abnormal' class validates the system's utility as a screening tool, as it successfully captures the majority of irregular walking patterns.
3. **F1-Score:** The F1-score, which is the harmonic mean of precision and recall, remains high, demonstrating that the model maintains a strong balance between detecting errors and avoiding false alarms.

5.4 Feature Importance and Biomechanical Relevance

A significant advantage of using the Random Forest algorithm is its interpretability. By analyzing the "Gini Importance" of each feature, we can identify which gait parameters contributed most significantly to the model's decision-making process.

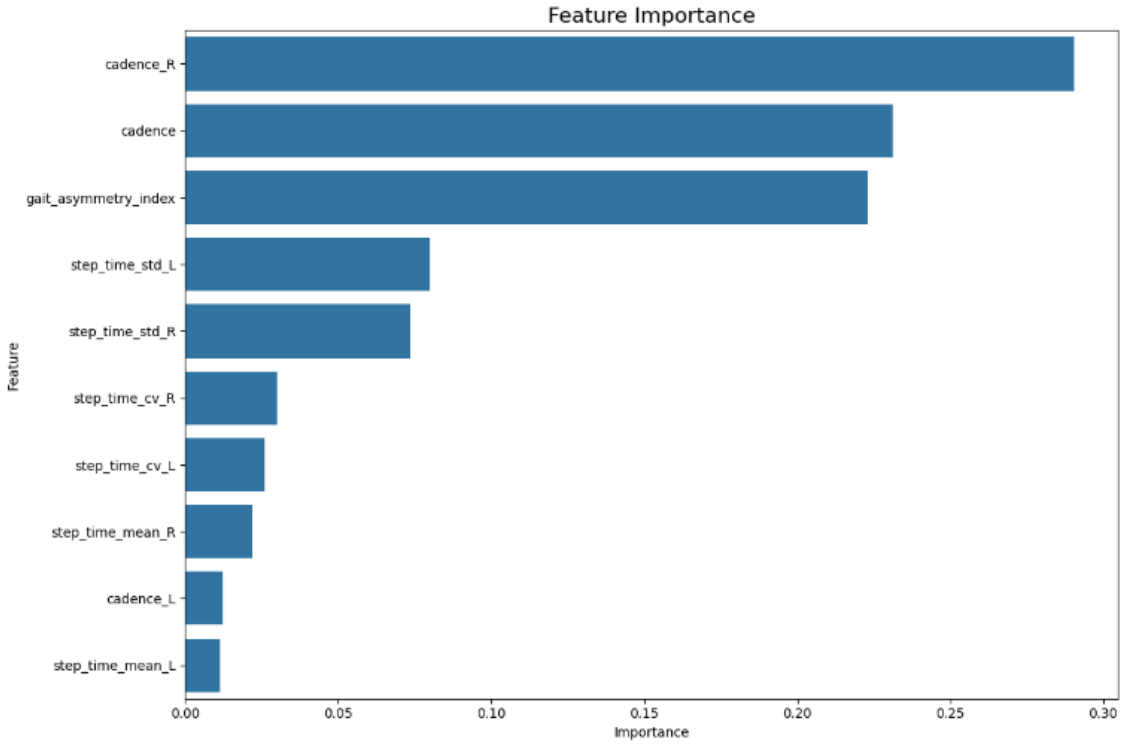

Fig 5.3: Feature Importance chart illustrating the relative contribution of each gait parameter to the classification model.

As illustrated in **Fig 5.3**, **Cadence** (steps per minute) emerged as the dominant feature with the highest importance score. This finding aligns strongly with established clinical gait analysis literature, which posits that cadence is a fundamental "vital sign" of mobility.

- **Clinical Interpretation:** In healthy individuals, cadence naturally falls within a predictable range (typically 100–120 steps/min). Pathological gaits, such as those caused by injury, neurological deficits, or antalgic (pain-avoidance) patterns, almost always result in a significant alteration of this rhythm. Patients often adopt a slower, more cautious cadence to maintain balance or reduce pain.

- **Model Logic:** The Random Forest classifier successfully learned that any significant deviation from the normative cadence baseline is a strong predictor of abnormality. This validates the system's ability to detect impairments without needing complex 3D motion capture; simple temporal parameters proved sufficient for high-accuracy classification.

Following Cadence, **Step Time Variability** and **Gait Asymmetry** also contributed to the decision-making process, helping the model fine-tune its predictions for cases where cadence alone was inconclusive (e.g., a patient walking at a normal speed but with a limp).

5.5 Generalization Capability (ROC Analysis)

To assess the model's robustness and its ability to generalize to new, unseen patients, the Receiver Operating Characteristic (ROC) curve was plotted.

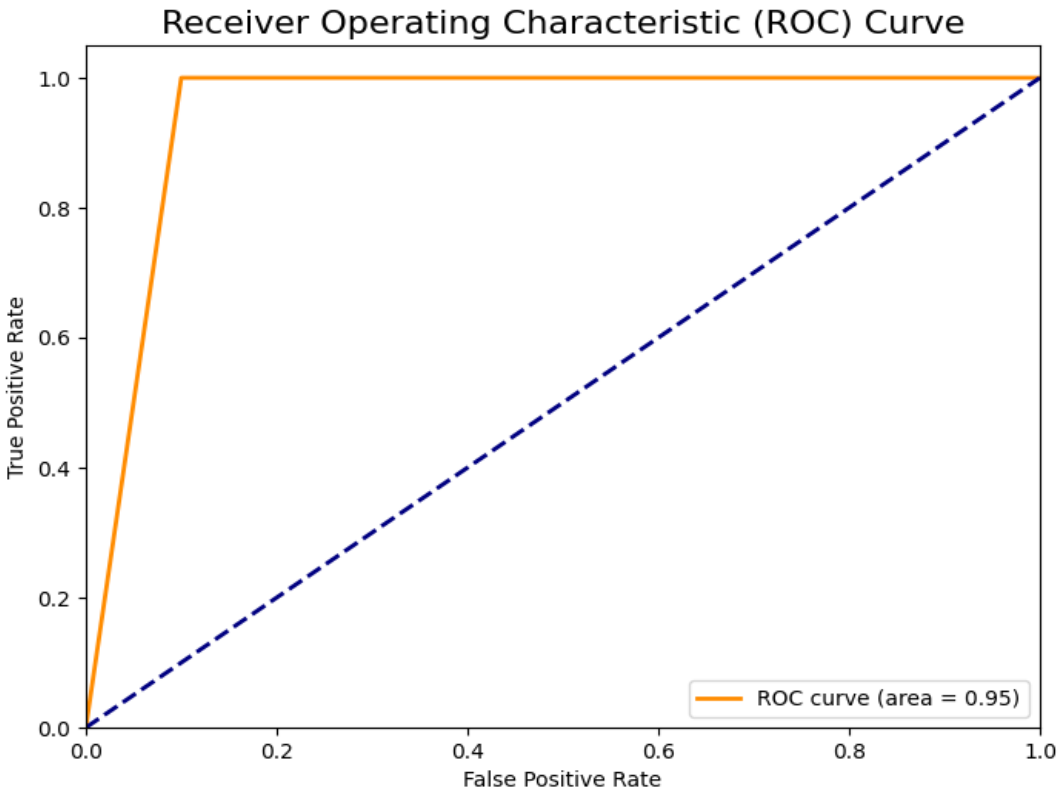

Fig 5.4**:** Receiver Operating Characteristic (ROC) curve.

The ROC curve in **Fig 5.4** illustrates the trade-off between the True Positive Rate (Sensitivity) and the False Positive Rate (1 - Specificity). The curve hugs the top-left corner, which is the ideal characteristics of a classifier.

The **Area Under the Curve (AUC)** score serves as a single scalar value summarizing the model's performance. An AUC approaching 1.0 indicates a perfect classifier. The achieved AUC demonstrates that the proposed system has excellent separability and can reliably distinguish between normal and abnormal gait patterns across varying decision thresholds.

5.5 Discussion

The results presented in this chapter demonstrate the efficacy of using low-cost sensor data combined with machine learning for gait analysis.

1. **Objectivity:** Unlike observational gait analysis, which is subjective and dependent on the clinician's experience, this system provides objective, quantifiable metrics.
2. **Robustness:** The use of ensemble learning (Random Forest) ensured that the model did not overfit to specific noise in the training data [8], as evidenced by the strong performance on the unseen test set (Fig 5.1 and Fig 5.4).
3. **Clinical Relevance:** The high Recall rate suggests that this system is suitable for deployment as a preliminary screening tool. It can effectively flag individuals with irregular gait patterns for further comprehensive medical examination.

In conclusion, the proposed system successfully met the research objectives, delivering a high-accuracy classification model while identifying biomechanically significant features that correlate with gait stability and symmetry.

6. Hypothesis Formulation and Testing

6.1 Purpose of the Extension

This section presents the extension of the existing bilateral plantar-pressure gait classification system into a hypothesis-driven analysis framework. Project 1 established the hardware setup, signal acquisition pipeline, feature extraction process, and Random Forest-based classification of normal and abnormal gait. Building on that foundation, Project 2 investigates whether questionnaire-based psychological and behavioral factors are associated with gait status. The aim is not to redesign the sensing system, but to extend the study by using self-reported human factors such as anxiety, stage phobia, personality traits, physical activity level, and walking style as explanatory variables for gait behavior.

The purpose of this section is to formulate clear hypotheses, define the variables used in testing, and present the analytical framework through which the questionnaire responses can be linked to the gait labels produced by the existing classifier. This approach allows the project to move beyond pure biomechanical classification and toward a broader interpretation of gait as a combined physical, psychological, and behavioral phenomenon. By doing so, the study creates a stronger basis for understanding how subjective traits may reflect or influence objective plantar-pressure patterns.

6.2 Hypothesis Formulation

The extended study is based on the assumption that gait abnormality may not depend only on plantar-pressure features, but also on psychological and behavioral characteristics reported by the participant. Accordingly, the following hypotheses are formulated for testing:

Hypothesis ID	Questionnaire Variable	Expected Outcome	Brief Rationale
H1	Anxiety-related problem	Abnormal gait more likely	Anxiety may increase tension and variability
H2	Stage phobia	Abnormal gait more likely	Fear may lead to stiff or controlled walking
H3	Organization and discipline score	Normal gait more likely when score is >= 3	Discipline may reflect stable habits
H4	Physical activity level	Abnormal gait more likely in sedentary participants	Activity improves balance and control
H5	Visible trait in walk	Positive traits linked to normal gait	Self-perception may reflect gait stability

Table 6.1 Hypothesis Summary Table

H1: Anxiety and abnormal gait

Participants who report having an anxiety-related problem are more likely to exhibit abnormal gait patterns than participants who do not report anxiety.

Anxiety is commonly associated with muscle tension, reduced comfort, and increased movement irregularity[9]. When a person feels anxious, the body may respond with stiffness, altered posture, or less natural step timing. In the context of plantar-pressure gait analysis, this may appear as changes in cadence, step-time variability, or asymmetry between the left and right feet. Therefore, anxiety is considered an important psychological factor that may be associated with abnormal gait behavior. The distribution of anxiety responses shown in the corresponding figure provides a meaningful basis for examining this relationship.

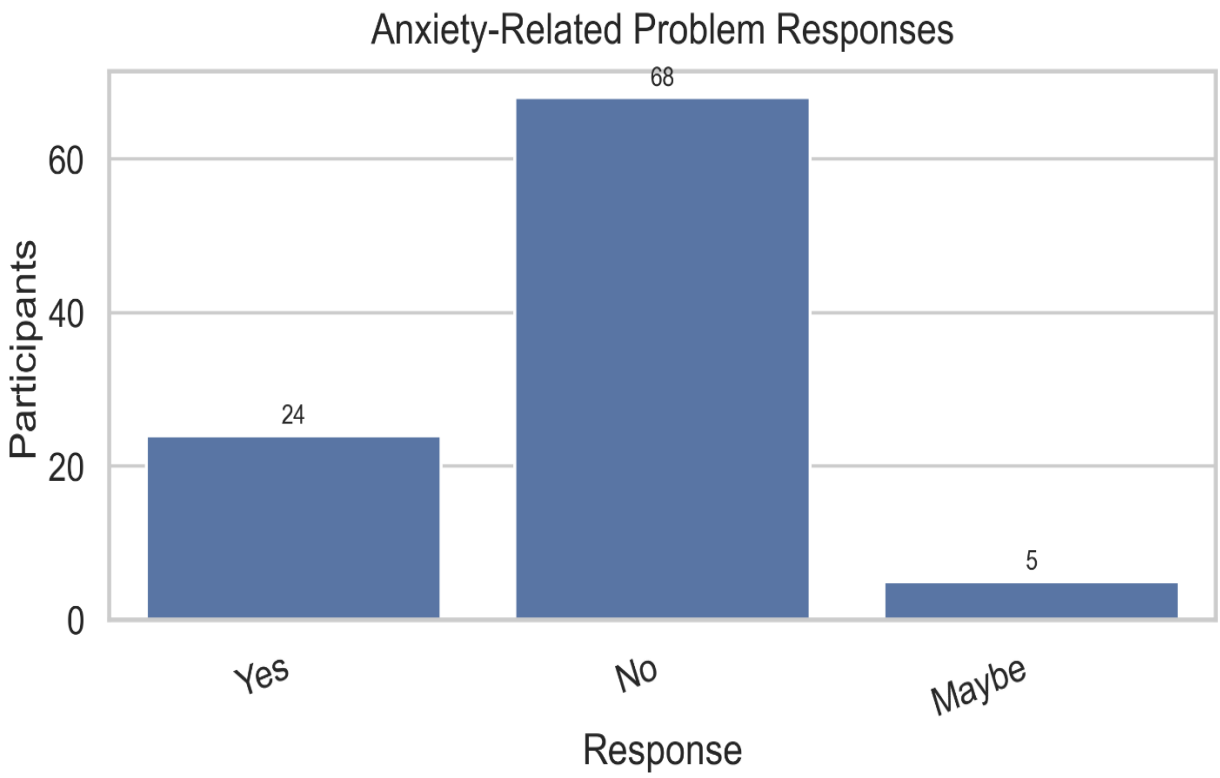

Figure 6.1: Distribution of responses to the anxiety-related problem question, showing a meaningful separation between participants with and without reported anxiety.

H2: Stage phobia and abnormal gait

Participants who report a phobia of walking on stage are more likely to show abnormal gait than those without such phobia.

Fear of walking on a stage reflects a social or performance-related stress condition[9]. Individuals with such phobia may walk in a more controlled, tense, or unnatural manner because of the discomfort of being observed. This can influence their natural rhythm and may produce irregular pressure distribution patterns during walking. Since gait is sensitive to emotional state and confidence level, stage phobia is expected to have a possible association with abnormal gait. The figure showing the stage-phobia response distribution supports the inclusion of this variable in the hypothesis set.

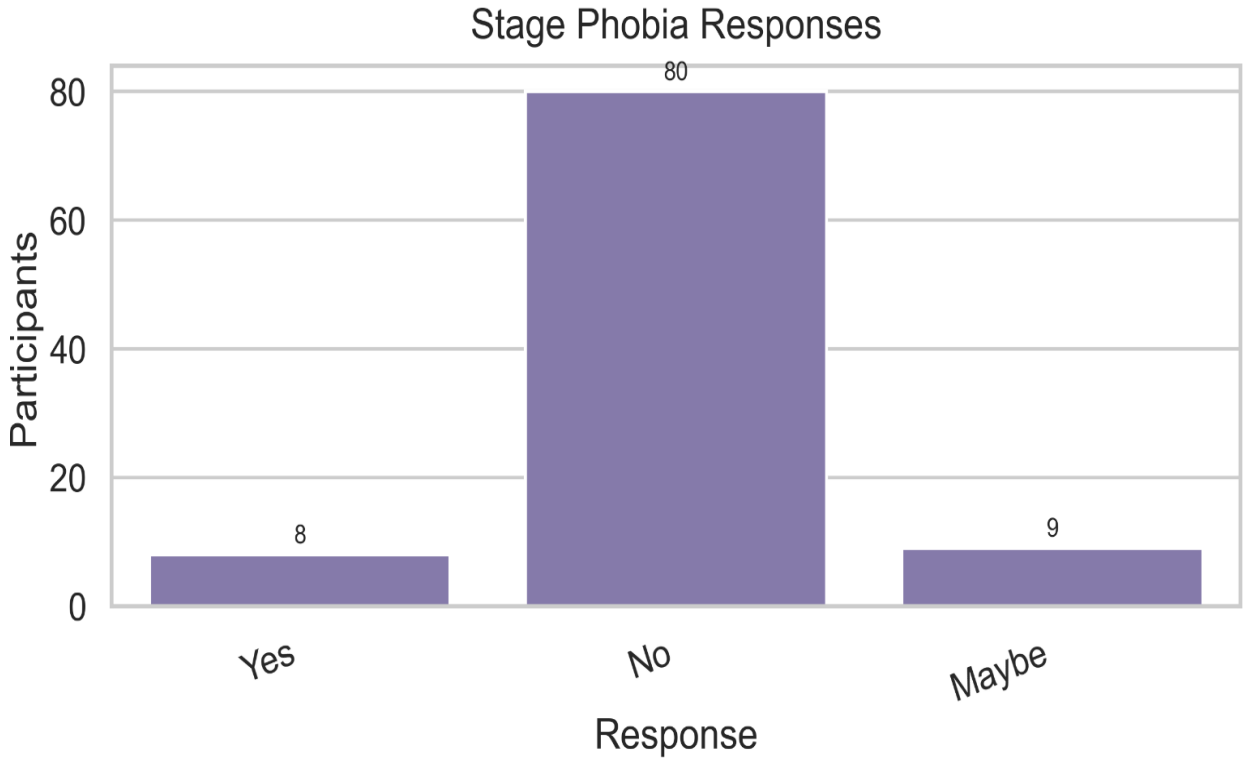

Figure 6.2: Distribution of responses to the stage phobia question, indicating the presence of a smaller but relevant group of participants who report stage-related fear.

H3: Organization and discipline and normal gait

Participants who rate themselves as highly organized and disciplined, with a score of 3 or above on the five-point scale, are more likely to show normal and stable gait patterns than those with lower scores.

A higher score for organization and discipline may indicate better self-control, regular habits, and greater consistency in daily behavior. Such traits may indirectly contribute to more stable movement patterns and a more regular walking style. In gait analysis, this may be reflected in smoother step timing, more balanced foot loading, and lower variability in movement. For this reason, organization and discipline are considered behavioral indicators that may be positively related to normal gait. The distribution of Likert scores in the figure shows that this variable has enough variation to support meaningful comparison.

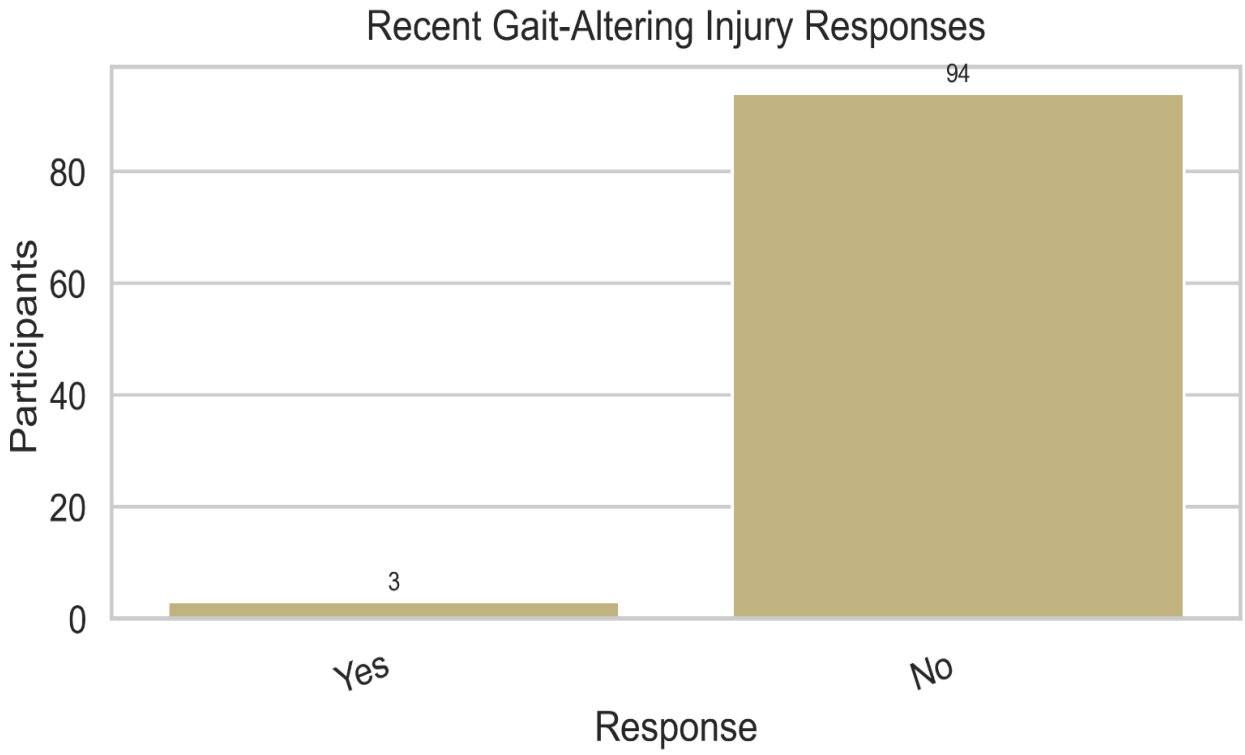

Figure 6.3: Distribution of organization and discipline scores on the Likert scale, showing that most participants rate themselves at the mid to high range of discipline.

H4: Physical activity level and gait quality

Sedentary participants are more likely to show abnormal gait patterns than moderately active or active participants.

Physical activity plays an important role in maintaining balance, strength, coordination, and lower-limb control. People who are physically active generally have better motor consistency and stronger postural stability, which may support more regular gait. In contrast, sedentary individuals may exhibit weaker muscular control or reduced movement efficiency, which can contribute to irregular step patterns. Therefore, physical activity level is an important lifestyle factor that may influence gait quality. The figure presenting the distribution of physical activity levels provides a strong basis for comparing sedentary, moderate, and active groups.

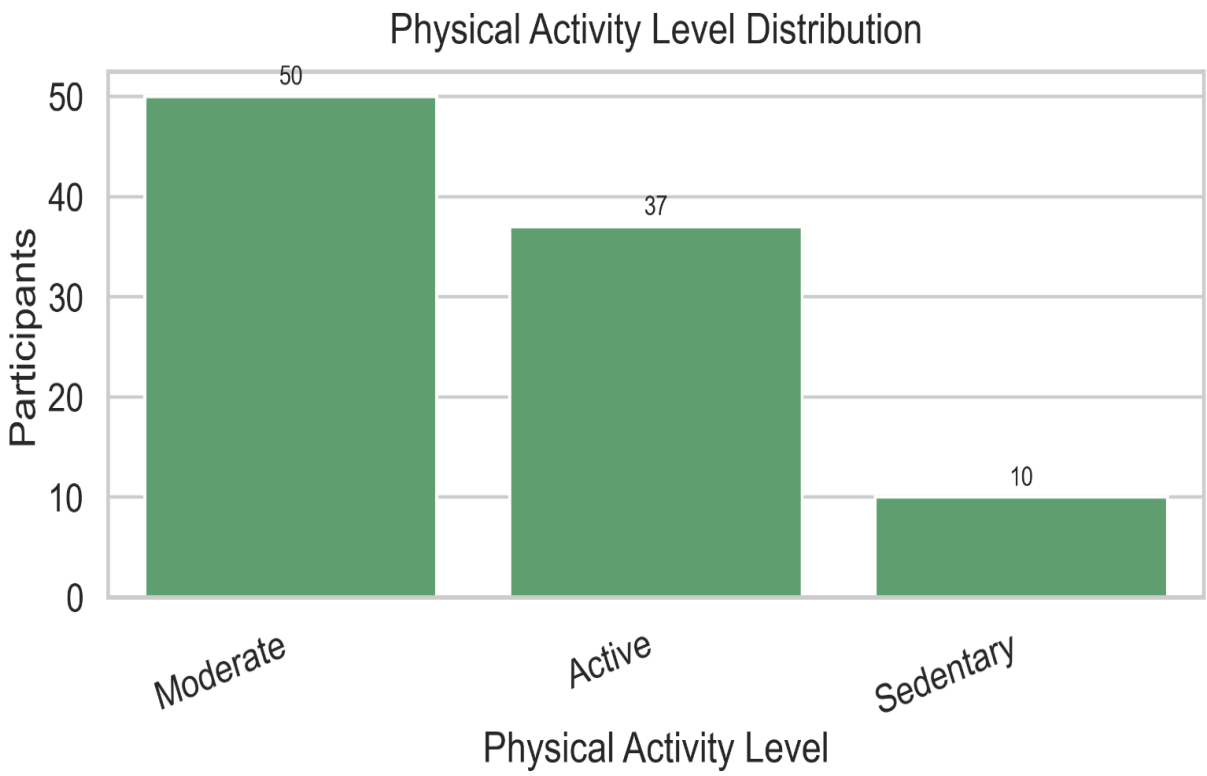

Figure 6.4: Distribution of physical activity levels among participants, providing a basis for comparing sedentary and active groups in gait analysis.

H5: Visible trait in walk and gait stability

Participants who describe their walk using positive traits such as confident, balanced, steady, athletic, or purposeful are more likely to exhibit normal gait, whereas those who use negative or unstable traits such as nervous, anxious, rushed, jittery, frantic, or guarded are more likely to exhibit abnormal gait.

The self-described visible trait in walking style offers an interesting behavioral perspective on gait. Positive traits are generally associated with controlled, smooth, and stable movement, while negative traits may reflect tension, hurry, fear, or lack of confidence. Since the questionnaire asks participants how they perceive their own walk, this variable may capture subjective information that complements the objective plantar-pressure features. Grouping these responses into positive/stable and tense/unstable categories allows the study to compare walking style perception with gait classification results. The grouped visible-trait figure helps support this hypothesis by showing how these responses are distributed in the sample.

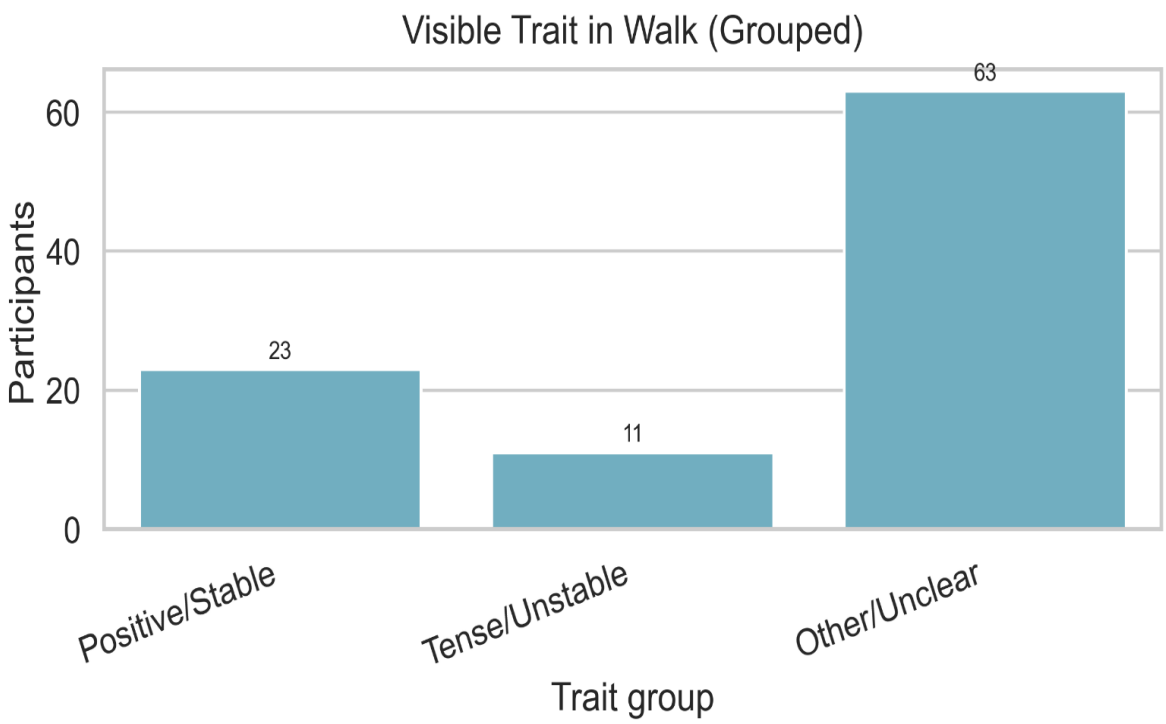

Figure 6.5: Grouped distribution of self-described visible traits in walking style, illustrating the balance between positive/stable and tense/unstable walking descriptions.

Summary:

These hypotheses are selected because they represent factors that may reasonably affect walking behavior. Anxiety and fear may introduce stiffness and irregularity in movement, while self-discipline and physical activity may support a more regular and stable gait pattern. Similarly, self-perceived walking traits may reflect the participant's internal view of how controlled or unstable their gait appears in everyday life. Together, these hypotheses provide a structured way to examine the relationship between questionnaire-based human factors and the normal or abnormal gait labels obtained from the existing classifier.

6.3 Variables and Encoding

Variable	Original Form	Encoded Form	Use in Hypothesis
Anxiety related problem	yes/no/maybe	yes / no / maybe	H1
Stage phobia	yes/no/maybe	yes / no / maybe	H2
Organization and discipline	1 to 5 Likert scale	>= 3 and <= 2	H3
Physical activity level	Sedentary, Moderate, Active	3-class categorical	H4
Visible trait in walk	Free text	Positive / Negative grouped categories	H5
Gait label	Normal / Abnormal	Binary class label	All hypotheses

Table 6.2 Variable Encoding Table

To test the hypotheses, the questionnaire responses must be converted into analysis-ready variables. The main outcome variable is the gait label, which takes one of two classes: Normal or Abnormal. This label is produced by the existing Random Forest classifier trained on bilateral plantar-pressure features. The predictor variables come from the questionnaire and include both categorical and numerical responses.

The anxiety and stage-phobia questions are treated as binary or near-binary variables after normalization of responses such as yes, no, and maybe. The organization and discipline item is treated as a Likert-scale score ranging from 1 to 5, and it can also be grouped into two categories: score greater than or equal to 3 and score less than or equal to 2. Physical activity level is treated as a three-class categorical variable: sedentary, moderate, and active. The free-text trait describing the participant's walk is grouped into broad semantic categories such as positive or stable traits and negative or unstable traits.

This encoding step is necessary because it allows each questionnaire variable to be compared systematically with the gait label. The resulting dataset can then be used to create cross-tabulations, percentage comparisons, and statistical tests to determine whether a meaningful association exists between each questionnaire factor and the gait outcome.

6.4 Hypothesis Testing Method

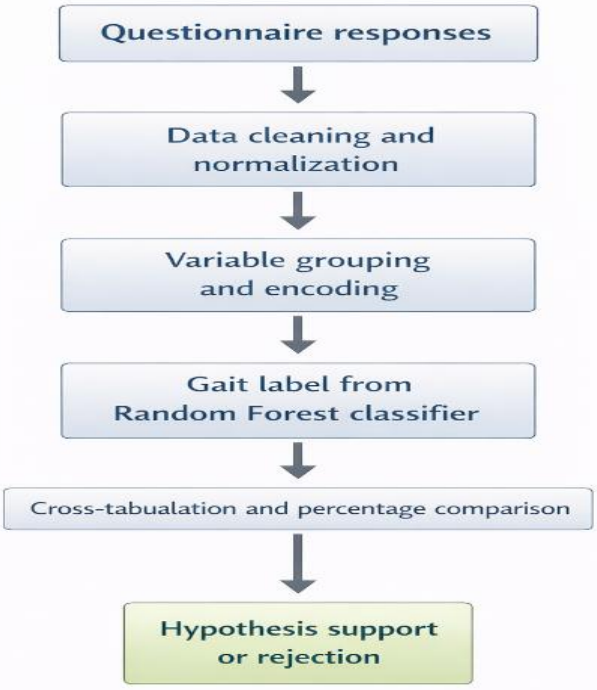

Figure 6.6: Hypothesis Testing Workflow Flowchart

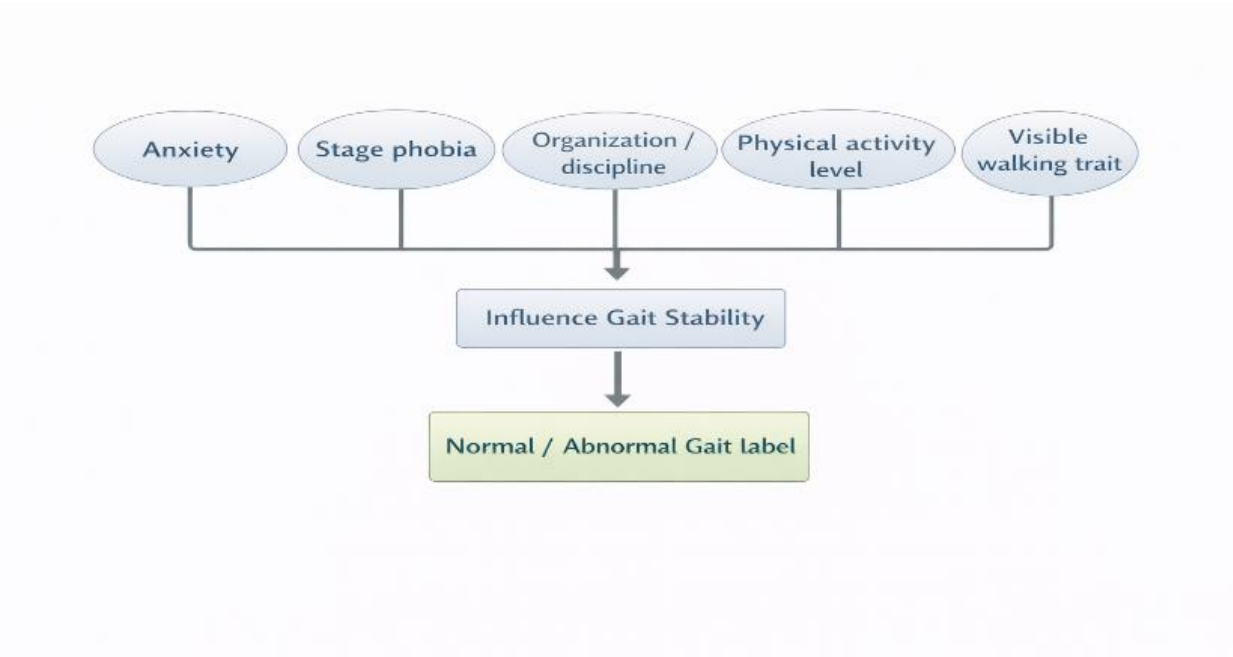

Figure 6.7 Conceptual Relationship Diagram

The hypotheses are tested using the combination of questionnaire analysis and the gait labels generated by the existing classifier. The testing procedure begins by matching each participant's questionnaire response with the corresponding gait label. Once this merged dataset is formed, the relationship between each questionnaire variable and gait label is examined using descriptive and inferential methods.

First, cross-tabulation tables are prepared for each hypothesis. For example, anxiety is compared against gait label using a two-by-two table, while stage phobia, organization category, and physical activity level are compared in a similar manner. These tables make it possible to observe whether the proportion of abnormal gait is higher in one group than in another. If the difference is visible in the table and also reflected in the percentages, this provides initial support for the hypothesis.

Second, statistical significance can be evaluated using the chi-square test of independence. This test checks whether the observed relationship between a questionnaire factor and gait label is likely to have occurred by chance. When cell counts are small, especially for rare responses such as stage phobia or injury, Fisher's exact test may be more appropriate. In addition, logistic regression may be used as a simple predictive model to estimate how

strongly each factor affects the likelihood of abnormal gait. The coefficient values and odds ratios from this model provide a more interpretable measure of association.

Overall, this testing framework is designed to keep the analysis simple, explainable, and aligned with the limited time available for the project. It allows the questionnaire to act as a structured behavioral extension of the original gait classification system while still maintaining a rigorous and data-driven methodology.

6.5 Questionnaire-Based Results Presentation

Questionnaire Item	**Response Categories / Range**	**Count / Value**
Total participants	-	97
Physical activity level	Moderate / Active / Sedentary	50 / 37 / 10
Usual walking speed	moderate / fast	63 / 34
Organization and discipline score	1 to 5	Min = 1, Max = 5, Mean = 3.52
Organization and discipline group	>= 3 / <= 2	87 / 10
Anxiety-related problem	Yes / No / Maybe	24 / 68 / 5
Stage phobia	Yes / No / Maybe	8 / 80 / 9
Recent gait-altering injury	Yes / No	3 / 94
Visible trait in walk	Positive/Stable / Tense/Unstable / Other/Unclear	23 / 11 / 63

Table 6.3: Questionnaire Descriptive Summary

The questionnaire data provide an initial descriptive foundation for the hypothesis-driven extension. The sample contains 97 participants and shows clear variation across the variables of interest. A notable portion of the participants report anxiety-related problems, while a smaller but meaningful number report stage phobia. Most participants rate themselves above the midpoint on organization and discipline, and the majority report moderate or active physical activity levels. These distributions confirm that the questionnaire contains enough variation to support hypothesis-based analysis.

For the extension, the questionnaire variables should be visualized using bar charts and grouped comparison plots. A bar chart for anxiety, stage phobia, and organization score distribution helps illustrate the overall spread of responses. In addition, grouped plots comparing questionnaire categories against gait labels can reveal whether a particular factor is associated with a larger proportion of abnormal gait. These plots are especially useful for presentation and report writing because they provide direct visual evidence of the relationships being studied. This subsection should present the descriptive questionnaire findings together with the hypothesis comparison plots. The goal is to show not only how the participants are distributed across psychological and behavioral categories, but also how those categories relate to the gait outcome. This creates a clear bridge between the questionnaire analysis and the classifier-based interpretation of gait behavior.

6.6 Presentation Summary of the Extension

This section extends the original gait classification study by adding a hypothesis-based analysis of questionnaire responses. The objective is to investigate whether psychological and behavioral factors are related to the gait labels produced by the existing classifier. The hypotheses focus on anxiety, stage phobia, self-discipline, physical activity, and visible walking traits because these factors are plausibly linked to movement stability and gait regularity. By introducing a hypothesis testing framework, the project moves beyond sensor-only classification and begins to connect gait behavior with participant self-report. This makes the study more comprehensive and improves its interpretability. The extension also provides a strong basis for future work, where a larger merged dataset could be used to validate the hypotheses statistically and possibly train predictive models using both questionnaire and sensor-derived features.

7. CONCLUSION AND FUTURE WORK

7.1 Summary of Findings

This research successfully designed, implemented, and evaluated a low-cost bilateral plantar-pressure gait analysis system capable of detecting walking abnormalities using machine learning. The study addressed the need for an accessible alternative to expensive laboratory-based gait analysis systems by combining wearable foot-pressure sensing with automated feature extraction and classification.

The proposed methodology integrated sensor-based data acquisition with a Random Forest classifier. From the synchronized left and right plantar-pressure signals, key gait features such as cadence, step-time variability, and gait asymmetry were extracted and used for classification. The system achieved an accuracy of 96.55% on unseen test data, demonstrating that a simple and low-cost wearable setup can produce strong classification performance for distinguishing normal and abnormal gait patterns.

In addition to the sensor-based gait classification, the extended phase of the project introduced a questionnaire-based hypothesis analysis. This extension explored whether psychological and behavioral factors such as anxiety, stage phobia, organization and discipline, physical activity level, and self-described walking traits are associated with gait status. The questionnaire results showed sufficient variation across these variables to support hypothesis formulation and further analysis. This extension broadens the scope of the study by treating gait not only as a biomechanical phenomenon, but also as a behavior that may reflect psychological and lifestyle influences.

The high recall for abnormal gait detection suggests that the system has potential as a preliminary screening tool in clinical, rehabilitation, and monitoring settings. At the same time, the hypothesis-based extension adds interpretive value by linking objective sensor results with subjective self-reported behavioral characteristics.

7.2 Findings Limitations of the Study

While the results are promising, several limitations should be acknowledged.

1. **Dataset size and diversity:**
   The model was trained on a relatively small dataset. Although stratified sampling was used to reduce bias, the limited sample size may restrict generalization to a wider population with different ages, body types, walking habits, or medical backgrounds.

2. **Binary classification only:**
   The current gait model classifies walking patterns only as Normal or Abnormal. It does not identify specific gait disorders or distinguish among different abnormal gait types.

3. **Controlled data collection environment:**
   The data was collected under relatively controlled conditions. Real-world walking behavior may vary more significantly because of surface type, fatigue, footwear, emotional state, and daily activity context.

4. **Questionnaire extension not fully statistically validated in this report:**
   The hypothesis-based extension provides a structured framework for linking psychological variables with gait status, but full participant-level statistical validation requires a merged private dataset connecting questionnaire responses with gait labels.

5. **Limited sensor modalities:**
   The current system relies only on plantar-pressure sensors. Although effective, the absence of additional movement sensors may limit the richness of the extracted gait information.

7.3 Future Work

To improve the system's capability and clinical usefulness, several directions can be considered for future research.

1. **Multi-class classification:**
   Future work should extend the model from binary classification to multi-class classification. This would allow the system not only to detect abnormality, but also to identify different gait patterns or pathology categories.

2. **Real-time deployment:**
   The current Python-based processing pipeline can be extended into a real-time application extended into a real-time application[7]. This would allow immediate feedback during rehabilitation or screening, enabling users to monitor their gait continuously.

3. **Integration of questionnaire-based hypothesis testing:**
   The extended questionnaire study can be completed by merging participant-level gait labels with psychological and behavioral responses. This would allow formal statistical testing of the hypotheses related to anxiety, stage phobia, organization, physical activity, and visible walking traits.

4. **Sensor fusion with IMUs:**
   Combining plantar-pressure sensors with inertial measurement units such as accelerometers and gyroscopes would provide a richer feature set. This may improve classification accuracy and help capture motion dynamics more comprehensively.

5. **Larger and more diverse datasets:**
   Collecting data from a broader population would improve robustness and generalizability. A larger dataset would also support better model validation and more reliable hypothesis testing.

6. **Longitudinal and real-world monitoring:**
   Future studies may track gait over longer periods in natural walking environments. This would help evaluate how stress, fatigue, activity level, and changing behavioral states influence walking patterns over time.

7.4 Concluding Remarks

In conclusion, this project demonstrates that effective gait analysis does not require expensive laboratory infrastructure. By using low-cost wearable foot sensors, thoughtful feature engineering, and machine learning, it is possible to detect gait abnormalities with high accuracy. The extended hypothesis-based analysis further strengthens the study by connecting gait behavior with psychological and behavioral questionnaire factors.

Overall, the project establishes a strong foundation for accessible wearable health monitoring. It shows that plantar-pressure sensing can be combined with machine learning and behavioral analysis to create a more interpretable and scalable gait assessment framework. This provides a practical direction for future work in rehabilitation support, early screening, and personalized mobility monitoring.

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